Placement Dataset Extension

Implementation plan + 28-CPU local smoke-test gate
Latent Causal Infeasibility for Long-Horizon TAMP | Finalized Approach B

Current authorization: inspect the repository, implement the smallest sidecar writer/physical trial adapter, run one tiny 2D serial check, then one bounded 28-worker local parallel smoke test. No full 5,000-root sweep, no 3D collection, no sampler/X/Y training, no end-to-end search.

Decisions frozen for this extension

Decision	Operational contract
Observed labels	Binary: verified placement = 1; attempted query with no witness before the pinned search cap = 0. Never call a finite-budget zero a universal physical impossibility proof.
Not attempted	No training label. A missing trial is absent/null and excluded from denominators. It is not a third observed class.
Grasp enumeration	Audit the declared pick/place mode identities (previous audit: 48 x 48). Continue after a first positive when collecting mode evidence, but full motion attempts are bounded.
Witnesses	Save each distinct verified path found, with actual joint/gripper/attachment/object histories and exact provenance.
Scene basis	Reuse selected 2D roots; new versioned sibling sidecar only. Hypothetical intermediate states are linked to parent root/split.
Compute	Local host only, at most 28 logical CPU workers; enforce aggregate CPU and memory/thermal protection; fail closed if preflight fails.

Authority and provenance. The 5 November 2026 Final-Fixed Approach B methodology controls the learned component split and the budgeted X/Y event definitions. The supplied 2D dataset audit reports 5,000 selected 2D roots, an 8 x 8 grid, nominal 48 pick and 48 place modes, plus existing .npz motion witnesses; Claude must re-check these claims against the actual checkout before coding. The smoke test creates new measured facts, not a paper result.

1. Scientific boundary: what is being labelled

The sampler predicts where an object that has already been grasped can plausibly be transported, released and safely finished. This is not the same as whether the robot can pick that object from the original scene; it is also not the same as executing the entire ordered correction or restoring the target task.

Quantity	Teacher question	Belongs to
Individual placement V_place	Given a declared post-grasp state for object o, can transport + placement/release + local completion at destination d be verified?	Sampler sidecar
Complete ordered X_r	Starting from original S0,q0, can the robot perform all correction moves in order, carry achieved states forward and connect terminally to q_ref?	Existing X data
Hypothetical Y_s	From fixed q_ref, does the original target task work in the final edited object arrangement?	Existing Y data
Trajectory witness	What actual robot joint, gripper, contact/attachment and object motion was verified?	Offline teacher evidence, not a sampler trajectory-generation target

1.1 Operational binary policy: no unresolved observed label

For an attempted eligible query, the label is exactly 1 or 0. A complete physical witness under the pinned protocol produces 1. If the declared object-destination trial budget is consumed without such a witness, record 0. A timeout/node cap/failed motion trial is a budget-policy-relative negative, not certified impossibility. Track the reason independently.

```
if not eligible: log_exclusion(reason); no placement trial
elif not attempted: label = null          # not in training loss
elif verified_witness_exists: label = 1
else: label = 0                      # bounded unsuccessful query
# Never replace "not attempted" with label 0.
```

A cell outside the legal lattice is static-ineligible and normally excluded from the learned conditional placement domain. If "out of bound" means planner resource exhaustion, that is an attempted operational 0. Preserve these two meanings separately.

1.2 No new architecture

The cVAE and flow-matching samplers will later learn distributions over destinations. Mode success ratios and generative density are not automatically calibrated success probabilities. No training occurs in this smoke. The main root-consistent joint X/Y Boltzmann model, cost ranking, branch-local resampling logic and original data remain untouched.

2. Exact collection unit and grasp-mode accounting

One trial key is (source root, hypothetical object-scene hash, object ID, source pose, destination pose, support, placement protocol, robot/profile hash, teacher budget hash). A change to any key component can change the observed event; do not silently deduplicate across different protocols.

2.1 Teacher mode enumeration

• For 2D, first verify the code actually declares 48 pick modes and 48 place modes. If so, the nominal pair catalogue is 2,304 per object/destination; record any permitted compatibility restrictions.

• Cache checks that are independent of the opposite end: valid post-grasp state, grasp transform, placement IK/geometry, static support and contact legality. Reject only by sound necessary checks.

• For each remaining compatible mode, run a real bounded full-3R transport/release/retreat check if it is selected within the smoke trial policy. Keep trying after the first verified success to measure diversity, stopping only at the pinned trial caps.

• A complete witness is the only positive evidence. A failed grasp mode is not a negative for every other mode, and a failure to pick from the initial scene is not a negative for the conditional placement event.

• Record explicit attempted masks for physical mode queries. "Listed in the 2,304-mode catalogue" is not equivalent to "attempted by the continuous planner".

2.2 Reconciliation fields

Per object-destination record	Meaning
modes_total	All nominal declared mode pairs; record exclusions/compatibility convention.
modes_geometrically_rejected	Sound cheap necessary-condition failures; includes invalid IK etc. with scope.
modes_motion_attempted	Pairs actually submitted to real bounded continuous motion verification.
modes_verified_success	Pairs with complete verified placement witnesses.
modes_budget_failed	Attempted pairs without a complete witness before their pinned cap.
modes_not_motion_tested	Remaining enumerated pairs not sent to the full planner; not a physical negative.

Require the count reconciliation invariant under the chosen mutually exclusive categories. Prefer an int8 outcome array and a bool attempted_mask over an oversized JSON line for each mode. If no complete witness exists after the declared query policy runs, the aggregate query may be operational 0 even when some nominal mode combinations were not physically explored. Report this policy limitation transparently.

2.3 Existing witness reuse

A stored original .npz trace is reusable as a new placement positive only if source/hypothetical scene, robot geometry/margins, grasp state, destination, transport/contact/release/retreat completion and replay policy all match. Otherwise preserve it as historical reference without relabeling. Do not change an archived first-witness travel label.

3. Intermediate states: part of the sidecar, not a new dataset

The 2D scene roots remain immutable. For a recorded parent prefix such as A to P, form the object-only edited state S^A and label a placement for a remaining object B. Such a state may be hypothetical even when no robot could execute A. This is legitimate sampler training context, provided provenance is explicit.

Case	Record as	Do not infer
Actual complete relocation prefix witnessed	parent_reachability = VERIFIED; link full source trajectory and achieved state	That every alternative B placement is feasible.
Object-pose edit only	parent_reachability = HYPOTHETICAL; store prefix+scene hashes	That the robot really achieved S^A or that an intermediate q_A is known.
B cannot initially be picked	Pick condition kept separate from V_place; B remains eligible for later sampler calls	Global exclusion of B or failure of A-then-B.
A alone does not restore target	X/Y may score A as poor stopping candidate	That AB and ABC should be pruned.

3.1 Minimal intermediate smoke requirement

Choose 1-2 intermediate contexts from the same selected roots if matching parent data exists. Include an example where the move changes occupancy or access for a later object. Save the original source hash, ordered parent prefix, full edited-scene hash, parent reachability flag and split. If none exists, report the limitation and do not fabricate an executed state.

3.2 Root grouping and leakage

Root ID is the group key for train/validation/test. Original scene, all descendant hypothetical scenes, all candidate destinations, order variants, trajectory witnesses and rerenders remain with that root. A smoke script must never regenerate root splits or use construction/intended blocker metadata as a sampler feature.

4. Minimal sidecar layout

```
dataset_bd_classquota/            # ORIGINAL: do not write

dataset_relocations_v1/
  smoke_<run_id>/
    manifest.json
    records/<root_id>.json
    modes/<root_id>.npz
    trajectories/<witness_hash>.npz
    visuals/<root_id>_placements.png
    visuals/<root_id>_witness.gif      # only for real witness
    summary.json
    report.md
```

The directory names above are proposed; inspect and match existing codebase conventions. Do not add a large package tree, database, separate trajectory-generating policy, or multiple duplicated teacher adapters.

5. Storage schema, round-trip checks and visualization

5.1 Human-readable record: conceptual example

```
{
  "schema_version":"placement_smoke_v1", "source_root_id":"<id>",
  "source_root_hash":"<hash>", "domain":"planar_2d",
  "prefix_key":[], "parent_reachability":"ROOT",
  "hypothetical_scene_hash":"<hash>",
  "object_id":"<o>", "destination_cell":37,
  "support_id":"table", "placement_protocol_id":"<hash>",
  "attempted":true, "label":0,
  "negative_kind":"BUDGET_EXHAUSTED", "is_certificate":false,
  "failed_phase":"CARRY", "witness_refs":[],
  "mode_summary":{"total":2304,"motion_attempted":8,
                  "verified_success":0},
  "budget_hash":"<hash>", "teacher_hash":"<hash>"
}
```

Example values above illustrate schema semantics only; they are not measured trial outcomes. A positive record uses label 1 and at least one independently verified trajectory witness reference. Missing/not-attempted records have label null or are absent from the observed set.

5.2 Compact arrays

File	Required payload
modes/.npz	Stable query-to-array index mapping, pick/place mode IDs, geometry checks, attempted_mask, observed binary mode outcomes, failure-phase codes, time/cost counters. Save array shapes/dtypes in manifest.
trajectories/.npz	Actual robot joint states/time, gripper aperture, attachment flags, object pose through phases, phase boundaries, start/end robot states, contact/replay provenance and measured travel. No numerical NaNs.
manifest.json / summary.json	Source Git version; root, robot, planner and budget hashes; split and seed; count reconciliation; wall and CPU resource observations; file relative paths, checksums and schema version.

5.3 Required visuals

• Placement map: actual source scene, support grid and full current object arrangement; green positive, red attempted operational 0, grey unattempted and hatched static-invalid.

• At least one saved real witness should have a phase-stepped filmstrip or GIF with articulated 3R links, fingers, attached object, and timestamps. Never substitute a translating-disc animation.

• The report must provide an image generated from the saved data after reopening the files, not a hand-drawn illustration as measurement evidence.

6. Tiny, deterministic local smoke schedule

Gate	Scope	Pass condition
G0: read-only preflight	Inspect actual repo/branch/configs, selected roots, source hashes, robot/witness API, local CPU/memory/swap/disk/thermal/affinity.	Report what exists. Fail/STOP on missing robot/teacher or fewer than 28 allowed logical CPUs if the 28-worker phase is required.
G1: single-worker sanity	One selected 2D root, one object, one legal destination; real teacher and save/reload.	Schema and arrays valid. A real positive trace must be verifiable before claiming witness-storage success.
G2: 28-worker bounded pass	At most 4 roots, 2 movable obstacles/root, 4-5 destinations/object: roughly 28-40 independent trial units plus 1-2 intermediate queries if available.	No worker leaks, bounded resource use, deterministic status accounting, reproducible output.
G3: round-trip + failure test	Reopen all JSON/NPZ/PNG/GIF. Repeat/resume idempotently and simulate interrupted worker.	No duplicate output, torn writes, mismatched hashes or spurious training labels.
G4: audit and STOP	Write report.md, summary.json and inventory.	SMOKE_PASS / SMOKE_FAIL / SMOKE_BLOCKED plus specific evidence. No following stage.

6.1 Candidate selection without hidden repair bias

Predeclare all root IDs, object IDs, and legal destination cells before running teacher outcomes. Cover near/far destinations and at least one destination not inherited from an intended repair. Prefer a source root with a known candidate witness for checking trace I/O but do not filter the entire smoke set on positive outcome. All selected trials must link to actual selected roots.

6.2 Small smoke search budgets

Start with at most 8 full-motion mode attempts per object/destination, at most 2 seconds per attempted mode, at most 12 seconds per object/destination and a 240-second global wall-time guard. Enumerating/checking all nominal pair identities is still allowed as a cheap bookkeeping step. These are newly versioned smoke-only policies, not changes to archived baseline/production budgets. If the existing teacher cannot enforce cancellation safely, STOP rather than leaving runaway processes.

Critical acceptance nuance. If every scheduled physical query returns operational 0 and no real positive trace is produced, the sidecar writer can be correct but the requested verified trajectory storage has not been demonstrated. Report SMOKE_BLOCKED or a scoped partial pass; do not fabricate a positive or silently expand to more roots.

7. Local resource policy: 28 logical CPUs, not 28 unlimited processes

The user authorizes a local 28-CPU smoke test, not a shared-server campaign. The run must use a preflight and an enforced upper bound. The hardware may expose 32 logical threads rather than 32 physical cores, and 28 busy threads can still generate substantial heat and memory pressure.

Resource	Required guard
CPU affinity	Read the allowed cpuset and pin the run to a logged set of 28 allowed logical CPU IDs. Never blindly assume IDs 0-27 are available.
Aggregate CPU	Prefer cgroup CPUQuota=2800% with max_workers=28. Affinity restricts where; CPU quota restricts aggregate CPU time. Nice alone is not a hard cap.
Threading	Set OMP_NUM_THREADS=1, MKL_NUM_THREADS=1, OPENBLAS_NUM_THREADS=1, NUMEXPR_NUM_THREADS=1 and configure planner backends to avoid nested pools.
Memory / swap	Measure per-worker peak RSS in serial preflight; only enable all 28 if estimated peak plus OS margin fits. Monitor MemAvailable, swap use and OOM; abort on unsafe pressure.
Thermal / disk	Log readable temperature sensors when available and enforce an explicit safety cutoff; guard low disk space and output size. If thermal readout is missing, warn and use short bounded workload.
Timeouts	Single global 240-s watchdog; bounded per-query and planner timeouts; terminate and join child processes on failure and interruption.

7.1 Example launch wrapper (adapt after repo inspection)

```
# Example only: replace command/path with inspected smoke CLI.
# CPUQuota requires a working user systemd CPU controller.
OMP_NUM_THREADS=1 MKL_NUM_THREADS=1 OPENBLAS_NUM_THREADS=1 \
NUMEXPR_NUM_THREADS=1 \
systemd-run --user --scope -p CPUQuota=2800% -- \
  nice -n 10 taskset -c "<28_VALID_ALLOWED_CPU_IDS>" \
  python placement_smoke.py --workers 28 --smoke-only

# If systemd quota is unavailable: report limitation;
# do not claim nice/taskset enforces aggregate CPU quota.
```

The wrapper is illustrative. Claude must print the exact selected affinity set and actual tested launch command. It must not use the SSH endpoint, attempt a remote deployment, or launch unbounded GPU jobs.

8. Required tests and stop conditions

8.1 Functional acceptance

• Historical dataset is byte-for-byte unchanged; archived replays and existing affected tests still work.

• Every attempted eligible trial has binary 1/0; no observed "UNRESOLVED" class. Missing/unattempted queries never count as zero.

• Every 1 has a complete matching independently verified witness; all cited trajectory files exist, reopen and satisfy phase/gripper/attachment/endpoint integrity.

• All per-mode counts reconcile and all actual motion attempts are distinguishable from geometric screens and not-tested pairs.

• Intermediate hypothetical scenes link to root, original split, ordered prefix and edited-scene hash; no fabricated achieved robot state.

• Rerunning deterministic output and resuming after a forced interrupt produces no duplicate witnesses or truncated JSON/NPZ.

• PNG/GIF correspond to saved physical data; at least one image visibly shows the correct real articulated robot geometry.

• No training, no new 3D roots, no cVAE/flow/X/Y changes, no background process after stop.

8.2 Mandatory report fields

The smoke report must include: repository/commit, config and teacher hashes; original root IDs and split; predeclared trial roster; full 48x48 catalogue verification; attempted/positive/zero/not-attempted counts; geometry and per-mode attempt count; witnesses reused vs newly verified; exact files and checksums; real trace lengths/phase boundaries; example failure reasons; source/edited scene plots; test suite result; resume/idempotence result; selected 28 CPU IDs and CPU quota enforcement; workers, peak RSS/MemAvailable, swap/disk/temperature readings, elapsed time; all limitations and STOP status.

Hard stop: After exactly one serial sanity test and one bounded 28-worker local smoke pass (with re-open/resume tests), stop. Do not start any larger scene coverage, add automatic retry campaigns, train a sampler, change teacher truth conventions, or run on the shared CPU-only server.

8.3 Explicitly deferred work

Production-scale placement labeling on 5,000 original 2D roots; conditional VAE and flow matching; feasibility-confidence calibration; sampler family comparisons; branch-local runtime integration; sampler-conditioned X/Y retraining; separate 3D labeling; downstream order/causal contrasts and held-out end-to-end verification. These are planned next stages, not smoke-test deliverables.

9. Source references and change boundary

Primary: Final-Fixed Methodology (5 Nov 2026), especially Sections 1, 5, 7-9, 11, 16 and Appendix A. Companion supplied audit: "Audit of your 2D dataset and proposed relocation extension" - reported selected roots, existing witnesses and 48x48 mode convention, to be rechecked in the actual repository. This addendum differs only in smoke implementation and operational binary labeling: attempted budget-exhausted queries are zeros with precise provenance; unattempted queries are not observed labels. The original Approach B responsibility split and X/Y model are unchanged.